Other 1PI Vertices

We can extend the definition of the vertex function from 3 points to n points given by: $iV_n(k_1, \dots, k_n)$

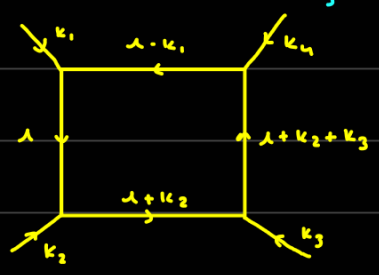
There are a few differences between $V_{n>3}$ and V_3 in the φ^3 theory:

① There is no tree-level contribution to $V_{n>3}$

② The one-loop correction to $V_{n>3}$ is finite for $d<6$.

This is important as the φ^4 theory operates in 6d.

As an example, we analyse the $n=4$ diagram:



This can be interpreted as:

$$iV_4 = g^4 \int \frac{d^6 l}{(2\pi)^6} \tilde{\delta}((l-k_1)^2)\, \tilde{\delta}((l+k_2)^2)\, \tilde{\delta}((l+k_2+k_3)^2)\, \tilde{\delta}(l^2) + O(g^6)$$

Thing is, there are multiple possible Feynman Diagrams here so we have to account for all of their contributions. $(k_2 \leftrightarrow k_3)$ and $(k_3 \leftrightarrow k_4)$

Q: What exactly is the vertex function and why is it a sum of all possible diagrams?

A: The vertex function is the sum of all 1PI diagrams with n external lines. By making it a sum and adding it to the Feynman Diagrams we are able to account for all possibilities from tree-level to many loop.

Using Feynman's formula for the propagators gives us:

$$\tilde{\delta}((l-k_1)^2)\, \tilde{\delta}((l+k_2))^2\, \tilde{\delta}((l+k_2+k_3)^2)\, \tilde{\delta}(l^2)$$

$$= \int dF_4 \left[x_1(l-k_1)^2 + x_2(l+k_2)^2 + x_3(l+k_2+k_3)^2 + x_4 l^2 + m^2\right]^{-4}$$

$$= \int dF_4 \left[q^2 + D_{1234}\right]^{-4} \qquad q = l - x_1k_1 + x_2k_2 + x_3(k_2+k_3) \; ; \; D_{1234} = x_1x_4k_1^2 + x_2x_4k_2^2 + x_2x_3k_3^2 + x_1x_3k_4^2 + m^2$$

$$+ x_1x_2(k_1+k_2)^2 + x_3x_4(k_2+k_3)^2$$

Now with a simple wick's rotation and application of the general gamma f^n result, we get:

$$\int \frac{d^6 q}{(2\pi)^6} \frac{1}{(q^2+D)^4} = \frac{i}{6(4\pi)^3 D}$$

Thus, we end up with:

$$V_4 = \frac{-g^4}{6(4\pi)^3} \int dF_4 \left(\frac{1}{D_{1234}} + \frac{1}{D_{1324}} + \frac{1}{D_{1243}} \right) + O(g^6)$$

This expression shows us that the 1-L contribution to V_4 is finite and well defined. The same

applies to U_{nsg}.